

**Definition 1.** Let  $P \in \mathbb{R}[x, y]$  be a squarefree polynomial, with  $\deg_y(P) > 0$ . We denote by  $\text{proj}(P)$  the set of points  $x \in \mathbb{R}$  such that either  $\text{Disc}_y(P)(x) = 0$  or  $\text{lc}_y(P)(x) = 0$ .

**Exercise 1.** Let  $P \in \mathbb{R}[x, y]$  be a squarefree polynomial, with  $\deg_y(P) > 0$ . Suppose that  $\#\text{proj}(P) < \infty$ . Let  $x_1, x_2, \dots, x_s$  be the elements of  $\text{proj}(P)$  in ascending order. For all  $i = 1, \dots, s$ , there exists a set  $A = \{y \in \mathbb{R} : P(x_i, y) = 0\}$ , and  $\#A = t$ . We can pose  $y'_j = (y_j + y_{j+1})/2$ . These lines help to divide the whole region into  $t$  regions. Prove that there exists  $\epsilon > 0$  such that every real branch of  $P = 0$  in the strip  $|x - x_i| < \epsilon$  is disjoint from all horizontal lines  $y = y'_j$ . Hence, near the vertical line  $x = x_i$ , each branch stays in a single horizontal region separated by the levels  $y'_1, \dots, y'_{t-1}$ .

*Proof.* For each  $x_i$ , fix  $j$ , we have  $P(x_i, y'_j) \neq 0$ . Pose  $f(x) = P(x, y'_j)$  is a polynomial function, so it's continuous. By the definition of continuity, there exists an  $\epsilon_j > 0$ , such that for all  $|x - x_i| < \epsilon_j$ , we have  $f(x) \neq 0$ , which means that the curve  $\forall x' \in (x_i - \epsilon_j, x_i + \epsilon_j), P(x', y'_j) \neq 0$ . Then for all  $y'_j$ , we can obtain an  $\epsilon_j$ . Let  $\epsilon = \min(\epsilon_j), j = 1, 2, \dots, t - 1$ , we have  $\forall x' \in (x_i - \epsilon, x_i + \epsilon), \forall j = 1, 2, \dots, t - 1, P(x', y'_j) \neq 0$   $\square$